KU LEUVEN BRUGGE

Analytical Evaluation of Magnetic Shielding Effectiveness of Infinite Planar Screens under Near-Field Conditions

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Thesis submitted for the degree of Master of Science in Engineering Technology, option Electronics and ICT

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Preface

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Albéric Simperl

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Abstract

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List of Abbreviations and Symbols

Abbreviations

EM	electromagnetic
EMC	electromagnetic compatability
SE	Shielding Effectiveness
RX	Receiving
TX	Transmitting
LF	Low frequency
NF	Near-field

Symbols

$J_n(.)$	Bessel function of the first kind and order n
$\boldsymbol{B}_i$	Magnetic field in medium i
$\boldsymbol{E}_i$	Electric field in medium i
$\boldsymbol{A}_i$	First order vector potential in medium i
$\boldsymbol{U}_i$	Second order vector potential in medium i

Chapter 1

Introduction

1.1 What is shielding?

Electromagnetic shielding is a means of improving electromagnetic compatibility (EMC) using an electromagnetic (EM) shield. Using this shield, it is possible to reduce the electromagnetic field intensity within a prescribed region [1]. A shield's effectiveness is denoted by its shielding effectiveness (SE), defined as the ratio of the field at a point of observation without and with the shield in place,

$$\mathrm{SE} = 20 \log_{10} \left| \frac{\boldsymbol{F}_{\mathrm{open}}}{\boldsymbol{F}_{\mathrm{closed}}} \right|, \ [\mathrm{dB}], \tag{1.1}$$

where $\boldsymbol{F}$ stands for either the electric or the magnetic field. For the magnetic problem considered in this work this becomes

$$\mathrm{SE}_H = 20 \log_{10} \left| \frac{H_{\mathrm{open}}}{H_{\mathrm{closed}}} \right|, \ [\mathrm{dB}]. \tag{1.2}$$

Depending on the material used, a shield can reflect, absorb or redirect the incident field. Shielding is used whenever a field source and a sensitive region must be isolated from each other. In military settings it can be used to guarantee communication security, protecting sensitive information transmitted using electromagnetic waves. Medical use cases are found in the isolation of MRI and ECG/EEG units, which need to be strictly enclosed. Furthermore, active implanted medical devices like pacemakers and cardioverters require an enclosure which protects these sensitive detection circuits. Although shielding is used in virtually any field, and whether this happens through immunity, emission control, privacy or even human safety, the objective is always ensuring EMC.

1.2 Low-frequency magnetic shielding

Not every field is equally easy to shield. At high frequencies, or far away from the source, the incident field behaves as a plane wave, in this case a thin metal sheet

already provides a large attenuation. At low frequencies the situation is different. Sources such as transformers, motors and power cables produce a magnetic field that changes slowly in time, and the objects that need protection are usually located close to them. The shield is then placed in the near field of the source.

In the near field of a current loop, the magnetic field dominates over the electric field. Such a source is called a low-impedance source. It is precisely this type of field that is hardest to stop, a metal sheet reflects almost nothing of it, and at low frequencies the sheet is often thinner than the depth the field reaches. A shield that blocks an electric field almost completely can let a large part of a low-frequency magnetic field pass. For this reason the low-frequency, low-impedance case is regarded as the worst case. Which is why it is the case that measurement standards use to qualify shielded rooms and enclosures.

These standards prescribe two loop antennas, a transmitting (TX) and a receiving (RX) loop, placed on either side of the shield. NSA 65-6 [2] places both loops on a common axis perpendicular to the shield (the coaxial configuration), while MIL-STD-285 [3] and its successor IEEE Std 299 [4] place them in a plane perpendicular to the shield (the coplanar configuration). The SE is obtained as the drop in the voltage induced in the RX loop when the shield is introduced. Because the RX loop has a finite size, this measured value is an average of the field over the loop area rather than the local field ratio of Equation (1.2). The SE that is measured and the SE that is defined are thus not the same quantity, and the difference depends on the dimensions of both loops and on their distance to the shield.

1.3 Problem statement and objective

Predicting the SE of a planar screen in this configuration is not straightforward. The field of a loop close to a shield is not uniform, so the shield is not illuminated by a single wave but by a distribution that depends on the loop radius and on the setup geometry. The exact solutions that exist take the form of integrals over Bessel functions, which give little direct insight in how each parameter influences the result, and the published results do not fully agree on how the SE depends on the distance between the loops.

This thesis develops and evaluates an analytical formulation for the magnetic SE of an infinite planar screen under near-field conditions, excited by finite loop antennas, within the quasi-static regime. The infinite planar screen is the reference geometry of both the literature and the standards, as it isolates the shielding mechanism from the effect of seams, apertures and edges. The aim is a model in which the influence of the shield thickness, conductivity and permeability, the loop radius and the spacing can be read directly, validated against the exact solutions in the literature.

Chapter 2 introduces the physics behind near-field magnetic shielding. Chapter 3 reviews the existing analytical formulations together with the expression for the SE that each provides, and identifies the open points that this work addresses.

Chapter 2

Physics behind near-field magnetic shielding

2.1 The quasi-static near-field regime

The boundary between the near- and far-field region of a small radiator lies at

$$r = \frac{\lambda}{2\pi} = \frac{1}{k}, \quad k = \omega\sqrt{\mu_0 \varepsilon_0}, \tag{2.3}$$

so that the near-field condition reads $kr \ll 1$. For frequencies from a few hundred hertz up to several tens of kilohertz this boundary lies between hundreds of kilometres and a few kilometres, so any laboratory set-up or practical enclosure operates deep inside the near field. Delay across the configuration is negligible, as the attenuation $e^{-jkr} \approx 1$, and the field at the shield is set by the instantaneous source current and by the geometry, not by the wavelength.

Inside a shield of conductivity σ the ratio of displacement to conduction current is $\omega\varepsilon/\sigma$, which for aluminium at 10 kHz is of the order of 10^{-18}. Dropping the displacement term turns the Helmholtz equation into a diffusion equation,

$$\nabla^2 \boldsymbol{H} = j\omega\mu\sigma \boldsymbol{H}, \quad \gamma = \sqrt{j\omega\mu\sigma} = \frac{1+j}{\delta}, \quad \delta = \sqrt{\frac{2}{\omega\mu\sigma}}, \tag{2.4}$$

with δ the skin depth. The field inside the shield does not propagate but diffuses, and the governing length scale is δ rather than λ.

The wave impedance of the source decides which kind of shielding is required. For a small loop and a short electric dipole the near-field wave impedances behave as

$$Z_w^{\text{loop}} \approx \eta_0 (kr), \quad Z_w^{\text{dipole}} \approx \frac{\eta_0}{kr}, \quad \eta_0 \approx 377\Omega. \tag{2.5}$$

Since $kr \ll 1$, a loop presents a low-impedance field and a dipole a high-impedance field, whereas both tend to η_0 in the far field.

2.2 Electric versus magnetic shields

For electric shielding the situation is comparatively trivial. The intrinsic impedance of a good conductor,

$$Z_s = \sqrt{\frac{j\omega\mu}{\sigma}} = \frac{1+j}{\sigma\delta}, \tag{2.6}$$

is of the order of micro-ohms at low frequency. Against the high-impedance field of Equation (2.5) this is an extreme mismatch, so an incident electric field is almost entirely reflected at the surface. Through charge redistribution in the conductor rather than diffusion through it, and the shield behaves as an equipotential surface.

The same mismatch is nearly absent for a low-impedance magnetic source, and reflection is weak. The shield must then act either by dissipating the field in induced currents or by offering the flux a low-reluctance path. These two mechanisms correspond to the two material classes discussed next.

2.3 Eddy-current shields

Electrically conductive shields, also called eddy-current shields, rely on currents induced by the incident time-varying magnetic field. This follows directly from Faraday's law,

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{2.7}$$

which states that a time-varying magnetic field induces an electric field. This field drives the free charges within the conductor, producing an induced current. When a magnetic field line falls onto a screen, the induced eddy currents form a closed loop around that field line, with a direction such that the induced field opposes the incident one, in accordance with Lenz's law. The shielded field is therefore the superposition of the incident field and the reaction field of the induced current distribution.

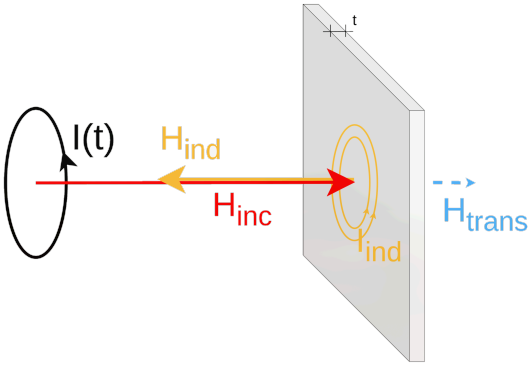


Figure 2.1: Eddy-current shielding principle.

2.4 Flux-shunting shields

Magnetically conductive shields, known as flux-shunting shields, are characterized by a high relative permeability μ_r to divert the flux into the shielding material. For an infinite planar screen this is most rigorously obtained from the jump conditions rather than from a lumped argument. Continuity of the normal flux density and of the tangential magnetic field,

$$B_{1n} = B_{2n}, \quad H_{1t} = H_{2t}, \tag{2.8}$$

combined with $\boldsymbol{B} = \mu \boldsymbol{H}$ in both media, gives the refraction law for the angles θ_1 and θ_2 measured from the interface normal,

$$\frac{\tan\theta_2}{\tan\theta_1} = \frac{\mu_2}{\mu_1} = \mu_r. \tag{2.9}$$

For $\mu_r \gg 1$, a field incident at any appreciable angle is refracted to $\theta_2 \to 90°$, meaning that the field inside the shield runs almost parallel to the sheet. Meaning the flux is drawn along the screen and guided away from the shielded region, which is what is observed macroscopically as a deflection of the field lines into the shield material.

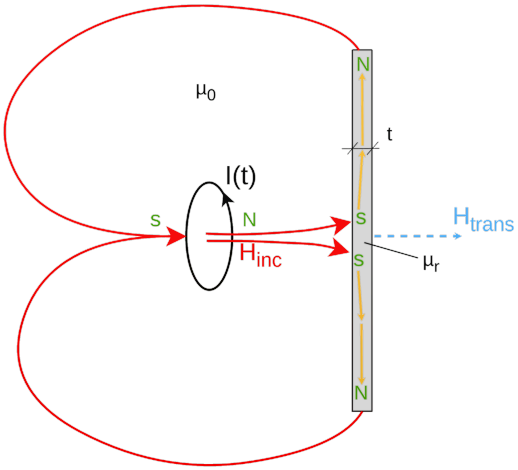


Figure 2.2: Flux-shunting shielding principle.

For closed or finite geometries the same behaviour is often described by the lumped magnetic-circuit analogue. A path of length l and cross-section A has reluctance $\mathcal{R} = l/(\mu_0 \mu_r A)$, and Hopkinson's law,

$$\Theta = \Phi \mathcal{R}, \tag{2.10}$$

relates the magnetomotive force Θ to the flux Φ, analogous to Ohm's law. As current over parallel resistors, the flux divides over parallel paths in inverse proportion to their reluctance, so that for $\mathcal{R}_{\text{shield}} \ll \mathcal{R}_{\text{air}}$ almost all flux follows the shield. It should be stressed that this description presupposes a well-defined flux path, which does not exist within an infinite planar sccreen. The reluctance has neither a unique l nor a unique A in that geometry. The magnetic-circuit picture is retained here as physical intuition only, while Equation (2.9) is the formulation used in the analysis.

2.5 Combined mechanisms and limits of the linear description

In practice the two mechanisms are not exclusive. A ferromagnetic shield of finite conductivity, such as steel, supports eddy currents and flux shunting simultaneously. At low frequency flux shunting dominates; as frequency rises the eddy-current contribution grows and takes over, so that a complete description must retain both effects through general μ and σ.

All of the above assumes a linear, homogeneous and isotropic medium. For ferromagnetic shields this holds only over a limited range: μ_r collapses towards unity near saturation, rolls off above a material-dependent frequency, and hysteresis adds a loss not captured by a real-valued μ_r. Closed-form results, including those used in this work, are therefore valid for small-signal excitation and below the permeability roll-off.

2.6 Characteristic length and time scales

Skin depth alone is not sufficient to predict the behaviour of a shield. For a screen of thickness t the governing dimensionless parameter is the ratio t/δ, which separates two distinct regimes.

For $t \gg \delta$ the field is attenuated exponentially inside of the material and absorption dominates. For $t \ll \delta$ the field is essentially uniform across the thickness, absorption vanishes, and the shielding is produced entirely by the field of the induced current. This thin-shield regime is the one in which thin metal sheets at low frequency operate, and it is the regime addressed by the classical eddy-current analyses discussed in Chapter 3.

In the thin-shield regime the shield has a characteristic diffusion time, which for a screen of thickness t at a distance d from the source scales as

$$\tau \sim \mu\sigma t d, \tag{2.11}$$

with the associated corner frequency $f_c \sim 1/(2\pi\tau)$. Below f_c the induced currents are too weak to react appreciably and the SE tends to 0 dB; above it the SE rises at approximately 20 dB per decade. Equivalently, the behaviour is governed by the dimensionless shielding parameter $\omega\mu\sigma t d$, which plays the role of a magnetic Reynolds number for the configuration. The proportionality in Equation (2.11) is a scaling relation; the exact prefactor depends on the geometry and follows from the closed-form solutions of Chapter 3.

Two consequences are worth noting. First, the corner frequency depends on the geometry through d, so the SE of a planar screen is not a material property alone. Second, since f_c scales inversely with $\mu\sigma t$, high-permeability and high-conductivity shields reach their useful range at lower frequencies, which is precisely why material choice is frequency-dependent.

2.7 The plane-wave spectrum of a loop source

The difference between near-field and far-field shielding lies in the spatial spectrum of the emitted field. As stated in the introduction, within the near field, we find the incident field to behave as a plane wave. This wave contains a single transverse wavenumber, whereas the near field of a finite loop contains a continuum of them.

Expanding the field of a circular loop of radius a in cylindrical coordinates yields a Sommerfeld-type integral over the transverse wavenumber λ, in which each spectral component carries a Bessel factor $J_1(\lambda a)$ set by the loop geometry and decays away from the source plane as $e^{\lambda\,|z|}$. In the quasi-static regime $\lambda \gg k$ for all components that carry appreciable energy, so the spectrum is essentially entirely evanescent. The dominant transverse wavenumbers are determined by the geometry, $\lambda \sim 1/a$ and $\lambda \sim 1/d$, and not by the frequency.

Each spectral component sees the shield at its own effective angle of incidence and is therefore transmitted with its own transmission coefficient $T(\lambda)$. The total shielded field is the superposition

$$\boldsymbol{H}_{\text{closed}}(\boldsymbol{r}) = \int_0^\infty T(\lambda)\, \widetilde{\boldsymbol{H}}(\lambda)\, e^{\lambda\,|z|}\, d\lambda, \tag{2.12}$$

so that the SE is an integral quantity weighted by the geometry of the source. Three conclusions follow directly, and they shape the remainder of this work. The SE of a planar screen necessarily depends on the loop radius a and on the spacings, since these set the spectral content. The Bessel-function integrands appearing in all of the closed-form solutions reviewed in Chapter 3 are the direct expression of this decomposition. And a model built on a single normally incident plane wave cannot be expected to reproduce the result, since it replaces a continuum of evanescent components by one propagating one.

Chapter 3

State of the art in the literature

The analytical treatment of low-frequency magnetic shielding by an infinite planar screen rests on a small and well-identified body of work, developed over roughly seven decades along two lines: thin-shield eddy-current analyses and exact solutions of the field equations. This chapter reviews them chronologically and gives, for each, its expression for the SE. The notation of the original papers is kept, so that the expressions can be checked against their source.

3.1 Levy's thin-shield analysis

The earliest treatment relevant to this configuration is the thin-shield eddy-current analysis of Levy [5],. Levy considered an infinite conducting sheet placed between two coaxial circular coils, formulated the problem in terms of magnetic potentials and satisfied the boundary conditions by postulating moving images. To make the problem tractable, the eddy currents are assumed to be distributed such that the induced potential is identical on both sides of the sheet, which restricts the result to shields thinner than a skin depth.

As summarised by Moser [6], combining the voltage induced in the receiving loop with and without the shield gives the insertion loss

$$S = 20\log_{10}\left|\frac{V_2''}{V_2}\right| = 10\log_{10}\left[\left(\frac{at\sigma_r f}{78.8}\right)^2 + 1\right], \tag{3.13}$$

in which a is the radius of the source loop and t the shield thickness, both in centimetres, σ_r the conductivity of the shield relative to that of copper, and f the frequency in hertz. V_2'' and V_2 are the voltages induced in the receiving loop without and with the shield.

Equation (3.13) describes a single corner frequency $f_c = 78.8/(at\sigma_r)$, below which the SE vanishes and above which it rises at 20 dB per decade, as anticipated by Equation (3.11). It is, however, not general. The expression is independent of the

loop spacing z and of the permeability of the shield, so it is restricted to thin, non-magnetic shields and to a limited frequency range. Moser reports that the omission of z is acceptable within ten percent only for $0.15 < r_1/a < 0.9$, with r_1 the distance from the source loop to the shield.

3.2 Moser's exact solution of the coaxial configuration

The work that defines the problem in its modern form is that of Moser [7], who analysed the shielding of a circular loop source by a plane metal sheet of infinite extent in the coaxial configuration of [2]. Besides Levy's analysis, Moser derived a direct solution of the vector wave equation, and measured aluminium, copper and steel shields between 100 Hz and 50 kHz.

Moser starts from the vector wave equation for the magnetic vector potential, $\nabla^2 \boldsymbol{A} + k_0^2 \boldsymbol{A} = 0$. For a loop of radius a carrying a current i the problem has cylindrical symmetry, so only the azimuthal component A_φ remains and $\nabla \cdot \boldsymbol{A} = 0$. Using the Fourier–Bessel integral, the free-space potential of the loop is written as

$$A_\varphi = \frac{\mu_0 a i}{2} \int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 \, |z|} d\lambda, \tag{3.14}$$

where the integration variable λ is the radial wavenumber, denoted k_ρ in Chapter 2.7 (and not the wavelength), and

$$\tau_0 = \sqrt{\lambda^2 - k_0^2}, \quad \tau = \sqrt{\lambda^2 - k^2}, \quad k^2 = j\omega\mu\sigma. \tag{3.15}$$

Equation (3.14) is precisely the spectral decomposition of Equation (3.12).

The shield, of thickness t, relative permeability μ_r and conductivity σ, occupies the region $b < z < c$, with b the distance from the source loop to the shield and $t = c - b$. Since the eddy currents in the plate have the same azimuthal symmetry as the source, the induced potential in each of the three regions (source side, shield, observation side) has the same form as Equation (3.14), multiplied by an unknown function of λ. These functions follow from the continuity of A_φ and of $(1/\mu)\partial A_\varphi/\partial z$ at $z = b$ and $z = c$. With $K = \tau/\tau_0$, the solution contains the common factor

$$C = \left[(K + \mu_r)^2 - (K - \mu_r)^2 e^{-2t\tau} \right]^{-1}. \tag{3.16}$$

Defining the SE as the ratio of the axial flux density without and with the shield, $S = 20 \log_{10} |B_z / B_z^{(3)}|$, consistent with Equation (3.2), and evaluating it on the axis ($r = 0$), Moser obtains

$$S = 20 \log_{10} \frac{1}{4\mu_r} \left| \frac{\int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a) e^{-\tau_0 z} d\lambda}{\int_0^\infty \frac{C\lambda^2 \tau}{\tau_0^2} J_1(\lambda a) e^{-\tau_0 z - t(\tau - \tau_0)} d\lambda} \right|, \tag{3.17}$$

with z the distance between the source loop and the point of observation. The SE is thus a quotient of two Sommerfeld-type integrals over $J_1(\lambda a)$. The numerator is the field of the unshielded loop; the denominator contains the full interaction with the shield through C and τ, and therefore both mechanisms of Chapter 2: eddy currents through σ and flux shunting through μ_r.

The 1967 paper did not evaluate Equation (3.17): the denominator has no known closed form, and numerical evaluation was unsuccessful at the time. In the quasi-static regime the numerator does have a closed form. For frequencies below 100 kHz, loop radii below 5 cm and $z < 10a$, the product $k_0\sqrt{a^2+z^2}$ is smaller than 10^{-3}, and the on-axis field of the unshielded loop reduces to the magnetostatic result

$$B_z = \frac{\mu_0 a^2 i}{2(a^2+z^2)^{\frac{3}{2}}}. \tag{3.18}$$

Substituting Equation (3.18) in Equation (3.17) gives

$$S = -20\log_{10}\left[\frac{4\mu_r}{a}(a^2+z^2)^{\frac{3}{2}}\left|\int_0^\infty \frac{C\lambda^2\tau}{\tau_0^2}J_1(\lambda a)e^{-\tau_0 z - t(\tau-\tau_0)}d\lambda\right|\right], \tag{3.19}$$

which leaves a single numerical integral. A notable property of Equation (3.17) and Equation (3.19) is that the source-to-shield distance b does not appear: the SE depends only on the total spacing z, and not on where the shield is placed between source and observation point.

In his 1988 treatment of the same geometry [6], Moser reports that Equation (3.17) has meanwhile been evaluated numerically by Ryan, with very favourable agreement with measurement, and that the independence of the source-to-shield spacing is consistent with the measured data.

3.3 Extension to loop-to-loop coupling

Equation (3.17) gives the SE at a point on the axis, whereas the standards measure the voltage induced in a finite receiving loop. Hansen and Moser [8] extended the formulation accordingly. Starting from Moser's off-axis result, the flux through a receiving loop of the same radius a is obtained by integrating over the loop area. With the change of variable $y = 1/\lambda$, which gives the integration variable the dimension of a distance, their expression reads (as published)

$$\mathrm{SE} = \frac{\frac{1}{4\mu_r}\int_0^\infty\int_0^a \frac{J_1(a/y)J_0(\rho/y)\exp(-\zeta_0 z/y)}{y^3\zeta_0}d\rho dy}{\int_0^\infty\int_0^a \frac{\zeta J_1(a/y)J_0(\rho/y)\exp[-\zeta_0(z-t)/y-\zeta t/y]}{y^3\left[(\zeta+\mu_r\zeta_0)^2-(\zeta-\mu_r\zeta_0)^2\exp(-2\zeta t/y)\right]}d\rho dy}, \tag{3.20}$$

with the normalised wavenumbers

$$\zeta^2 = 1 - j\omega\mu_0\mu_r\sigma y^2, \quad \zeta_0^2 = 1 - k^2y^2, \tag{3.21}$$

where here $k = 2\pi/\lambda_0$ is the free-space wavenumber and z the distance between the loops. Comparing with Equation (3.17), the integration over the receiving loop introduces exactly one additional Bessel factor $J_0(\rho/y)$ in both integrands, as Equation (3.12) would suggest. The authors note that the quasi-static closed form of the numerator, Equation (3.18), may give large errors once the receiving loop is included and the spacing is comparable to or smaller than the loop radius, so that both integrals have to be evaluated numerically.

The consequence of this small change is substantial. The on-axis result Equation (3.17), evaluated as a function of the spacing, shows a "bathtub" shape with a minimum of the SE for spacings comparable to the loop radius. With the receiving loop integration this anomaly disappears: the loop-to-loop SE varies monotonically with spacing and tends to 0 dB as the spacing tends to zero. The on-axis and loop-to-loop results only coincide when the spacing exceeds the loop radius.

This is a genuine discrepancy within the literature rather than a refinement of it, and it bears directly on the independence of the SE of the source-to-shield spacing reported in [6]. It is one of the points against which the formulation developed in this work is tested.

3.4 Exact treatment of the coplanar configuration

Araneo and Celozzi [9] addressed the coplanar loop configuration prescribed by [4] and [3]. Exact solutions had previously been found only for the coaxial geometry of [2]. Their solution accounts for the finite size of both the TX and the RX antenna, rather than using a small-dipole approximation, and is validated experimentally.

Since the coplanar configuration lacks the rotational symmetry of the coaxial one, the problem is formulated in Cartesian coordinates with the magnetic vector potential of the second order $\boldsymbol{U}$, defined by

$$\nabla \times \boldsymbol{U} = \boldsymbol{A}, \quad \nabla \times (\nabla \times \boldsymbol{U}) = \boldsymbol{B}, \tag{3.22}$$

and gauged such that $\nabla \cdot \boldsymbol{U} = 0$, so that $\nabla^2 \boldsymbol{U} = 0$ in the quasi-static limit and the field component of interest follows as $B_y = \partial^2 U_y / \partial y^2$. Two assumptions are made explicit: retardation is neglected, $e^{-jkr} \approx 1$, which holds for the IEEE Std 299 loop of diameter 30.5 cm up to 16 MHz, and the current in the source loop is uniform along its circumference. In the absence of the shield, the potential of a TX loop of radius R, carrying an RMS current I at a minimum distance d from the shield, is

$$U_y(x, y, z) = \int_0^\infty \int_0^\infty G_0(\alpha, \beta) \cos(\alpha x) \cos(\beta y) e^{-D_1 \, |z+d|} d\alpha d\beta, \tag{3.23}$$

$$G_0 = -\frac{\mu_0 I R}{\pi} \frac{1}{\beta D_1} I_1(\beta R), \quad D_1 = \sqrt{\alpha^2 + \beta^2}, \tag{3.24}$$

where $I_1(\cdot)$ is the modified Bessel function of the first kind and order one. When the shield of thickness t is applied, G_0 is replaced by

$$G_1 = -\frac{4\mu_0 I R}{\pi}\frac{D_3 e^{-(D_2-D_1)t}}{D_4 \beta} I_1(\beta R), \tag{3.25}$$

with

$$D_2 = \sqrt{\alpha^2+\beta^2+j\omega\mu\sigma}, \quad D_3 = \frac{D_2}{\mu_r}, \quad D_4 = (D_3+D_1)^2 - (D_3-D_1)^2 e^{-2D_2 t} \tag{3.26}$$

The ratio $G_1/G_0 = 4D_1 D_3 e^{-(D_2-D_1)t}/D_4$ plays the same role as the factor C in Equation (3.16): it is the transmission coefficient of the shield for a single spectral component (α, β).

The SE at a point of observation becomes

$$\mathrm{SE} = 20\log_{10}\left|\frac{H_y^i}{H_y^t}\right| = 20\log_{10}\left|\frac{\int_0^\infty\int_0^\infty \beta^2 G_0(\alpha,\beta)\cos(\alpha x)\cos(\beta y)e^{-D_1\,|z+d|}d\alpha d\beta}{\int_0^\infty\int_0^\infty \beta^2 G_1(\alpha,\beta)\cos(\alpha x)\cos(\beta y)e^{-D_1\,|z+d|}d\alpha d\beta}\right| \tag{3.27}$$

where the superscripts i and t denote the field without and with the screen. The integrands are smooth, and the double integrals are readily computed numerically. Unlike Equation (3.17), the result depends on the position of the shield between the two loops, since it depends on both d and the distance z from the RX antenna.

Since the meter reading is proportional to the flux linked with the RX loop, IEEE Std 299 defines the SE as $20\log_{10}(\Phi^i/\Phi^t)$. Araneo and Celozzi approximate the flux integral over the RX surface by a seven-point quadrature of B_y, and find that it differs by less than 5% from the SE at the centre of the loop for the cases analysed. They further show that the small-dipole approximation, assumed without proof in several textbooks and handbooks, is not justified when the loop dimensions are comparable to the spacings, as in the standardised set-ups.

The monograph of Celozzi, Araneo and Lovat [1] subsequently consolidated this material, and is used throughout this work as the general reference frame rather than as a primary source.

3.5 Overview of the formulations

Table 3.1 summarises the formulations reviewed above. All exact solutions share the same structure: the SE is a quotient of spectral integrals in which the shield enters only through a transmission factor per spectral component, while the geometry of the loops enters through Bessel factors.

Work	Config.	Observation	Shield	Form of SE
Levy [5]	coaxial	loop voltage	thin, $\mu_r = 1$	closed form, Equation (3.13)
Moser [6], [7]	coaxial	point on axis	general μ_r, σ, t	single integral, Equation (3.17)
Hansen & Moser [8]	coaxial	RX loop flux	general μ_r, σ, t	double integral, Equation (3.20)
Araneo & Celozzi [9]	coplanar	point / RX loop	general μ_r, σ, t	double integral, Equation (3.27)

Table 3.1: Analytical formulations for the SE of an infinite planar screen with loop sources.

3.6 Open points and position of this work

Taken together, the literature establishes exact solutions for the coplanar configuration with finite antennas [9], an evaluated exact solution for the coaxial configuration with a point of observation [6], and an extension of the latter to loop-to-loop coupling [8] that contradicts the earlier result on its dependence on spacing. Three gaps remain.

The dependence of the SE on the loop spacing in the coaxial configuration is reported differently in [6] and [8], and the two results have not been reconciled.

The small-dipole assumption, whose limits were quantified for the coplanar configuration in [9], has not been assessed to the same degree for the coaxial configuration with finite loops.

Finally, the available exact formulations are presented as evaluated integrals for specific material and geometry combinations, rather than as a parametrised model from which the dependence on each geometrical and material parameter can be extracted directly.

This work addresses these points by developing and evaluating an analytical formulation for the magnetic SE of an infinite planar screen under near-field conditions, with finite loop antennas, within the quasi-static regime delimited by Equation (3.3) and Equation (3.5), and by comparing it against the published results above.

Chapter 4

Generalisation to an arbitrarily placed receiving loop

The closed-form results of Chapter 3 give the field behind the shield only in the coaxial configuration, observed either in a point on the axis [6], [7] or with a receiving loop that is coaxial with and parallel to the source [8]. In this chapter the receiving loop is given an arbitrary position and orientation. No new boundary-value problem has to be solved for this: the axisymmetric solution behind the shield is retained, and only the receiving loop is moved by a rigid transformation. The flux it captures then follows from a single line integral.

The only result taken from the literature is Moser's axisymmetric vector potential behind the shield. Everything built on it in this chapter is the contribution of this thesis:

- the Cartesian and coordinate-free form of that potential, including its regular behaviour on the axis;
- the description of the receiving loop as a rigid transformation in SE(3);
- the flux through a loop in an arbitrary pose, reduced via Stokes' theorem to one integral over the loop;
- the invariances that reduce the six pose parameters to four;
- an orientation parametrisation based on unit quaternions, replacing Euler angles;
- the condition under which the result is valid;
- the SE for an arbitrary pose, and the poses for which it is not defined.

4.1 Assumptions and geometry

The global frame $(O; \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$ is attached to the transmitting loop, a filamentary circular loop of radius a carrying a current i, centred in O in the plane $z = 0$. The shield is the slab $b \leq z \leq c$ of Chapter 3, of thickness $t = c - b$, so that region 3 is the half-space $z > c$. The receiving loop is a filamentary circular loop of radius R. The analysis rests on four assumptions.

1. The quasi-static conditions of Chapter 2 hold, so that $\tau_0 = \lambda$ in Equation (4.15).

2. The receiving loop is open-circuited, or loaded by a high impedance, so that its own current perturbs neither the source nor the eddy currents in the shield. Its open-circuit voltage is then $V = -j\omega\Phi$, with Φ the flux through the loop.
3. The receiving loop lies entirely in region 3.
4. The shield is of infinite transverse extent, so that source and shield together remain axisymmetric about the z-axis.

4.2 Vector potential behind the shield

Under these assumptions the field in region 3 is the one derived by Moser [7]. Denoting the cylindrical radius by ρ, to keep $\boldsymbol{r}$ free for the position vector, it reads

$$\boldsymbol{A}^{(3)} = A_\varphi(\rho, z)\,\hat{\boldsymbol{\varphi}}, \quad A_\varphi(\rho, z) = 2\mu a i \int_0^\infty \frac{C\lambda\tau}{\tau_0^2} J_1(\lambda a) J_1(\lambda\rho) e^{-\tau_0 z - t(\tau-\tau_0)}\, d\lambda \tag{4.28}$$

with $\mu = \mu_0\mu_r$, τ and τ_0 from Equation (4.15) and C from Equation (4.16). Without the shield, A_φ is given by Equation (4.14) with r replaced by ρ. The derivation below only uses that $\boldsymbol{A}$ is azimuthal and axisymmetric, and therefore holds for both cases.

4.3 Unit vector transform

To evaluate Equation (4.28) in a Cartesian point (x, y, z), both the argument ρ and the unit vector $\hat{\boldsymbol{\varphi}}$ have to be expressed in x and y. The starting point is the radial unit vector, which points from the axis to the point under the angle φ with the x-axis, see Figure 4.1. Since $\cos\varphi = x/\rho$ and $\sin\varphi = y/\rho$,

$$\begin{aligned} \hat{\boldsymbol{\rho}} &= \cos\varphi\,\hat{\boldsymbol{x}} + \sin\varphi\,\hat{\boldsymbol{y}} \\ &= \frac{x}{\rho}\,\hat{\boldsymbol{x}} + \frac{y}{\rho}\,\hat{\boldsymbol{y}}. \end{aligned} \tag{4.29}$$

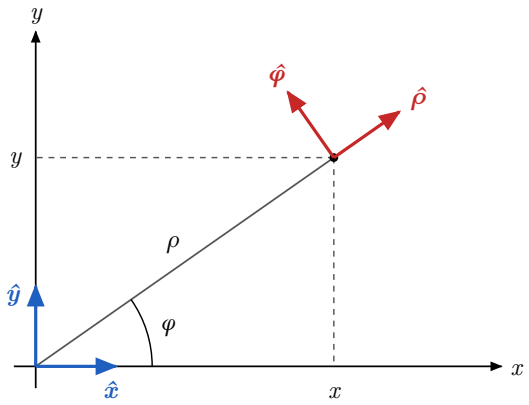


Figure 4.1: Carthesian and cylindrical unit vectors in a point in the xy-plane

The azimuthal unit vector lies in the same plane and is orthogonal to the radial one, $\hat{\boldsymbol{\rho}} \perp \hat{\boldsymbol{\varphi}}$, in the direction of increasing φ. It is therefore the radial unit vector rotated over $+\pi/2$:

$$\begin{aligned}
\hat{\boldsymbol{\varphi}} &= \cos\!\left(\varphi + \frac{\pi}{2}\right) \hat{\boldsymbol{x}} + \sin\!\left(\varphi + \frac{\pi}{2}\right) \hat{\boldsymbol{y}} \\
&= -\sin\varphi\, \hat{\boldsymbol{x}} + \cos\varphi\, \hat{\boldsymbol{y}} \\
&= -\frac{y}{\rho}\, \hat{\boldsymbol{x}} + \frac{x}{\rho}\, \hat{\boldsymbol{y}} \\
&= \frac{-y}{\sqrt{x^2+y^2}}\, \hat{\boldsymbol{x}} + \frac{x}{\sqrt{x^2+y^2}}\, \hat{\boldsymbol{y}}.
\end{aligned} \tag{4.30}$$

The vector potential behind the shield is purely azimuthal, $\boldsymbol{A}^{(3)} = A_\varphi(\rho, z)\, \hat{\boldsymbol{\varphi}}$. Substituting $\rho = \sqrt{x^2+y^2}$ in the argument and Equation (4.30) for the unit vector gives

$$\begin{aligned}
\boldsymbol{A}^{(3)}(x,y,z) &= A_\varphi(\rho, z)\, \hat{\boldsymbol{\varphi}} \\
&= A_\varphi\!\left(\sqrt{x^2+y^2}, z\right) \left[\frac{-y}{\sqrt{x^2+y^2}}\, \hat{\boldsymbol{x}} + \frac{x}{\sqrt{x^2+y^2}}\, \hat{\boldsymbol{y}} \right].
\end{aligned} \tag{4.31}$$

4.4 Flux through a laterally displaced loop

Equation (4.31) is first applied to a receiving loop of radius R that is parallel to the source but displaced from its axis, with centre (x_c, y_c, z_c). The flux through the loop follows from Stokes' theorem,

$$\Phi = \iint_S \boldsymbol{B} \cdot d\boldsymbol{S} = \iint_S (\nabla \times \boldsymbol{A}) \cdot d\boldsymbol{S} = \oint_{\partial S} \boldsymbol{A} \cdot d\boldsymbol{r}. \tag{4.32}$$

The loop is parametrised by the angle $\alpha \in [0, 2\pi)$ around its own centre,

$$\begin{aligned}
\boldsymbol{r}(\alpha) &= x(\alpha)\, \hat{\boldsymbol{x}} + y(\alpha)\, \hat{\boldsymbol{y}} + z_c \hat{\boldsymbol{z}} \\
&= (x_c + R\cos\alpha)\, \hat{\boldsymbol{x}} + (y_c + R\sin\alpha)\, \hat{\boldsymbol{y}} + z_c \hat{\boldsymbol{z}}, \\
d\boldsymbol{r} &= R(-\sin\alpha\, \hat{\boldsymbol{x}} + \cos\alpha\, \hat{\boldsymbol{y}})\, d\alpha.
\end{aligned} \tag{4.33}$$

On the loop, the distance to the axis of the source depends on α,

$$\rho(\alpha) = \sqrt{x^2(\alpha) + y^2(\alpha)} = \sqrt{(x_c + R\cos\alpha)^2 + (y_c + R\sin\alpha)^2}, \tag{4.34}$$

and so does the azimuthal unit vector, $\hat{\boldsymbol{\varphi}}(\alpha) = (-y(\alpha)\, \hat{\boldsymbol{x}} + x(\alpha)\, \hat{\boldsymbol{y}})/\rho(\alpha)$. Note that $\hat{\boldsymbol{\varphi}}(\alpha)$ differs from the direction $-\sin\alpha\, \hat{\boldsymbol{x}} + \cos\alpha\, \hat{\boldsymbol{y}}$ of $d\boldsymbol{r}$ as soon as the loop is displaced; only their dot product enters the flux. Working it out,

$$
\begin{aligned}
\boldsymbol{A}^{(3)} \cdot d\boldsymbol{r} &= A_\varphi(\rho(\alpha), z_c)\, \hat{\boldsymbol{\varphi}}(\alpha) \cdot d\boldsymbol{r} \\
&= A_\varphi\Big(\sqrt{x^2(\alpha) + y^2(\alpha)}, z_c\Big) \left(\frac{-y(\alpha)}{\rho(\alpha)}\, \hat{\boldsymbol{x}} + \frac{x(\alpha)}{\rho(\alpha)}\, \hat{\boldsymbol{y}} \right) \cdot R(-\sin\alpha\, \hat{\boldsymbol{x}} + \cos\alpha\, \hat{\boldsymbol{y}})\, d\alpha \\
&= A_\varphi(\rho(\alpha), z_c)\, \frac{R}{\rho(\alpha)} \left(y(\alpha) \sin\alpha + x(\alpha) \cos\alpha \right) d\alpha \\
&= A_\varphi(\rho(\alpha), z_c)\, \frac{R}{\rho(\alpha)} \left((y_c + R\sin\alpha) \sin\alpha + (x_c + R\cos\alpha) \cos\alpha \right) d\alpha \\
&= A_\varphi(\rho(\alpha), z_c)\, \frac{R}{\rho(\alpha)} \left(y_c \sin\alpha + R\sin^2\alpha + x_c \cos\alpha + R\cos^2\alpha \right) d\alpha \\
&= A_\varphi(\rho(\alpha), z_c)\, \frac{R}{\rho(\alpha)} \left(y_c \sin\alpha + x_c \cos\alpha + R \right) d\alpha.
\end{aligned}
\tag{4.35}
$$

Substituting in Equation (4.32) gives the flux through the displaced loop,

$$
\Phi = \int_0^{2\pi} A_\varphi(\rho(\alpha), z_c)\, \frac{R(x_c \cos\alpha + y_c \sin\alpha + R)}{\sqrt{(x_c + R\cos\alpha)^2 + (y_c + R\sin\alpha)^2}}\, d\alpha. \tag{4.36}
$$

With A_φ from Equation (4.28), written as $A_\varphi(\rho, z) = \int_0^\infty f(\lambda, z) J_1(\lambda\rho)\, d\lambda$ with $f(\lambda, z) = 2\mu a i C \lambda \tau / \tau_0^2\, J_1(\lambda a) e^{-\tau_0 z - t(\tau - \tau_0)}$, this becomes the double integral

$$
\Phi = \int_0^{2\pi} \int_0^\infty f(\lambda, z_c)\, J_1(\lambda\rho(\alpha))\, \frac{R(x_c \cos\alpha + y_c \sin\alpha + R)}{\sqrt{(x_c + R\cos\alpha)^2 + (y_c + R\sin\alpha)^2}}, \tag{4.37}
$$

4.5 Pose of the receiving loop as a rigid transformation

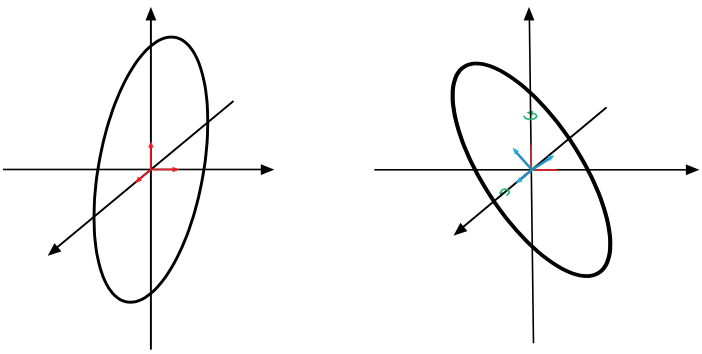


Figure 4.2: Transmitting loop of radius a in the plane $z = 0$ and receiving loop of radius R with centre $\boldsymbol{c} = z_c \hat{\boldsymbol{z}}$ and local frame $(\hat{\boldsymbol{u}}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{n}})$, tilted over θ with respect to $\hat{\boldsymbol{z}}$.

A local frame $(\boldsymbol{c}; \hat{\boldsymbol{u}}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{n}})$ is attached to the receiving loop, with $\boldsymbol{c} = (x_c, y_c, z_c)$ its centre and $\hat{\boldsymbol{u}}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{n}} = \hat{\boldsymbol{u}} \times \hat{\boldsymbol{v}}$ a right-handed orthonormal basis, $\hat{\boldsymbol{n}}$ being the loop normal, see Figure 4.2. In this frame the loop is the circle

$$\boldsymbol{p}(\alpha) = R(\cos\alpha, \sin\alpha, 0)^T, \quad \alpha \in [0, 2\pi), \tag{4.38}$$

whose sense of traversal fixes, by the right-hand rule, the positive direction of $\hat{\boldsymbol{n}}$ and therefore the sign of Φ and V. The local frame is mapped onto the global one by the affine transformation

$$T : \mathbb{R}^3 \to \mathbb{R}^3 : \boldsymbol{p} \mapsto \boldsymbol{c} + \boldsymbol{Q}\boldsymbol{p}, \quad \boldsymbol{Q} = (\hat{\boldsymbol{u}} \;\; \hat{\boldsymbol{v}} \;\; \hat{\boldsymbol{n}}) \in \mathrm{SO}(3), \tag{4.39}$$

or, in homogeneous coordinates,

$$\begin{pmatrix} \boldsymbol{r} \\ 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{Q} & \boldsymbol{c} \\ \boldsymbol{0}^T & 1 \end{pmatrix} \begin{pmatrix} \boldsymbol{p} \\ 1 \end{pmatrix}. \tag{4.40}$$

Since $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}$ and $\det \boldsymbol{Q} = 1$, T is a proper rigid motion, an element of the special Euclidean group SE(3). It preserves lengths, angles and orientation, so the image of the circle is again a circle of radius R, traversed in the same sense with respect to its normal. Only the linear part $\boldsymbol{Q}$ acts on vectors such as $d\boldsymbol{r}$; the translation $\boldsymbol{c}$ acts on points only. In global coordinates the loop becomes

$$\boldsymbol{r}(\alpha) = \boldsymbol{c} + R\hat{\boldsymbol{e}}(\alpha), \quad \frac{d\boldsymbol{r}}{d\alpha} = R\hat{\boldsymbol{t}}(\alpha), \tag{4.41}$$

with the radial and tangential unit vectors

$$\hat{\boldsymbol{e}}(\alpha) = \cos\alpha\,\hat{\boldsymbol{u}} + \sin\alpha\,\hat{\boldsymbol{v}}, \quad \hat{\boldsymbol{t}}(\alpha) = -\sin\alpha\,\hat{\boldsymbol{u}} + \cos\alpha\,\hat{\boldsymbol{v}}, \quad \hat{\boldsymbol{e}} \times \hat{\boldsymbol{t}} = \hat{\boldsymbol{n}}. \tag{4.42}$$

4.6 Flux through the receiving loop

The flux through the receiving loop again follows from Stokes' theorem, Equation (4.32), where S may be taken as the flat disc bounded by the loop. Region 3 is a half-space and therefore convex, so the disc lies in region 3 together with its boundary, and Equation (4.28) is valid on all of S. Because $\nabla \cdot \boldsymbol{B} = 0$, the result does not depend on the chosen spanning surface, nor on the gauge of $\boldsymbol{A}$.

Substituting x and using the cyclic property of the scalar triple product, $(\hat{\boldsymbol{z}} \times \boldsymbol{r}) \cdot d\boldsymbol{r} = \hat{\boldsymbol{z}} \cdot (\boldsymbol{r} \times d\boldsymbol{r})$. With Equation (4.41) and Equation (4.42),

$$\boldsymbol{r} \times d\boldsymbol{r} = (\boldsymbol{c} + R\hat{\boldsymbol{e}}) \times R\hat{\boldsymbol{t}}\, d\alpha = R\big(\boldsymbol{c} \times \hat{\boldsymbol{t}} + R\hat{\boldsymbol{n}}\big)\, d\alpha, \tag{4.43}$$

so that

$$\boldsymbol{A} \cdot d\boldsymbol{r} = g(\rho, z)\,(\hat{\boldsymbol{z}} \times \boldsymbol{r}) \cdot d\boldsymbol{r} = g(\rho, z)\, R\big[Rn_z + \hat{\boldsymbol{z}} \cdot \big(\boldsymbol{c} \times \hat{\boldsymbol{t}}\big)\big]\, d\alpha, \tag{4.44}$$

and the flux reduces to the single integral

$$\Phi = R\int_0^{2\pi} g(\rho(\alpha), z(\alpha))\big[Rn_z + \hat{\boldsymbol{z}}\cdot\big(\boldsymbol{c}\times\hat{\boldsymbol{t}}(\alpha)\big)\big]\,d\alpha, \tag{4.45}$$

with $n_z = \hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{n}}$ and

$$\rho(\alpha) = |\hat{\boldsymbol{z}}\times\boldsymbol{r}(\alpha)|, \quad z(\alpha) = \hat{\boldsymbol{z}}\cdot\boldsymbol{r}(\alpha) = z_c + R(u_z\cos\alpha + v_z\sin\alpha). \tag{4.46}$$

In terms of the components of $\boldsymbol{c}$ and of the columns of $\boldsymbol{Q}$,

$$\begin{aligned}\hat{\boldsymbol{z}}\cdot\big(\boldsymbol{c}\times\hat{\boldsymbol{t}}\big) &= x_c\big(v_y\cos\alpha - u_y\sin\alpha\big) - y_c(v_x\cos\alpha - u_x\sin\alpha),\\ \rho^2(\alpha) &= \big(x_c + R\,e_x(\alpha)\big)^2 + \big(y_c + R\,e_y(\alpha)\big)^2, \quad e_k(\alpha) = u_k\cos\alpha + v_k\sin\alpha.\end{aligned} \tag{4.47}$$

Equation (4.45) separates the two ways in which the pose enters. The term Rn_z stems from the projection of the loop onto the plane of the shield, the term in $\boldsymbol{c}$ from the offset of the loop with respect to the axis of the source. For a uniform field $\boldsymbol{B} = B_0\hat{\boldsymbol{z}}$, $g = B_0/2$ is constant, the offset term integrates to zero because $\int_0^{2\pi}\hat{\boldsymbol{t}}\,d\alpha = \boldsymbol{0}$, and Equation (4.45) returns $\Phi = \pi R^2 B_0 n_z$, as it should.

4.7 Reduction to known configurations

The configurations treated in the literature are special cases of Equation (4.45), which provides a first check of the formulation.

Coaxial, parallel loops. For $\boldsymbol{c} = z_c\hat{\boldsymbol{z}}$ and $\boldsymbol{Q} = \boldsymbol{I}$, one has $\rho(\alpha) = R$ and $z(\alpha) = z_c$, while $\boldsymbol{c}\times\hat{\boldsymbol{t}}$ is perpendicular to $\hat{\boldsymbol{z}}$. Equation (4.45) becomes $\Phi = 2\pi R A_\varphi(R, z_c)$, the loop-to-loop coupling of [8].

Parallel, laterally displaced loops. For $\boldsymbol{Q} = \boldsymbol{I}$, so that $\hat{\boldsymbol{u}} = \hat{\boldsymbol{x}}$, $\hat{\boldsymbol{v}} = \hat{\boldsymbol{y}}$ and $n_z = 1$, the offset term becomes $x_c\cos\alpha + y_c\sin\alpha$ and Equation (4.45) reduces to

$$\Phi = R\int_0^{2\pi} g(\rho(\alpha), z_c)(R + x_c\cos\alpha + y_c\sin\alpha)\,d\alpha, \tag{4.48}$$

which, with $g = A_\varphi/\rho$, is exactly Equation (4.36) of Chapter 4.4.

Small receiving loop. For $R\to 0$ the flux density is uniform over the disc and $\Phi\to\pi R^2\,\hat{\boldsymbol{n}}\cdot\boldsymbol{B}^{(3)}(\boldsymbol{c})$. On the axis and with $\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}$, the ratio of the open and shielded flux reduces to the point-wise SE of Moser, Equation (4.17).

4.8 Invariances and degrees of freedom

A rigid motion has six degrees of freedom, three in $\boldsymbol{c}$ and three in $\boldsymbol{Q}$. Not all of them affect Φ.

Rotation about the loop normal. Replacing $\boldsymbol{Q}$ by $\boldsymbol{Q}\boldsymbol{R}_z(\delta)$, with $\boldsymbol{R}_z(\delta)$ a rotation over δ about the local z-axis, maps $\hat{\boldsymbol{e}}(\alpha)\to\hat{\boldsymbol{e}}(\alpha+\delta)$ and $\hat{\boldsymbol{t}}(\alpha)\to\hat{\boldsymbol{t}}(\alpha+\delta)$ and leaves

$\hat{\boldsymbol{n}}$ unchanged. The new parametrisation describes the same closed curve in the same sense, shifted in α, and since the integrand of Equation (4.45) is 2π-periodic, Φ is unchanged. The flux therefore depends on $\boldsymbol{Q}$ only through $\hat{\boldsymbol{n}} = \boldsymbol{Q}\hat{\boldsymbol{z}}$: the relevant orientations form the quotient $\mathrm{SO}(3)/\,\mathrm{SO}(2) \cong S^2$, the unit sphere of loop normals.

Rotation about the axis of the source. Since source and shield are axisymmetric, $\boldsymbol{A}(\boldsymbol{R}_z(\gamma)\boldsymbol{r}) = \boldsymbol{R}_z(\gamma)\boldsymbol{A}(\boldsymbol{r})$ for every rotation about the global z-axis. Rotating the complete receiving loop, $\boldsymbol{c} \to \boldsymbol{R}_z(\gamma)\boldsymbol{c}$ and $\boldsymbol{Q} \to \boldsymbol{R}_z(\gamma)\boldsymbol{Q}$, leaves n_z, $\rho(\alpha)$, $z(\alpha)$ and $\hat{\boldsymbol{z}} \cdot \big(\boldsymbol{c} \times \hat{\boldsymbol{t}}\big)$ invariant, and hence Φ.

The flux is therefore a function of four parameters only: the radial offset $\rho_c = \sqrt{x_c^2 + y_c^2}$ and the height z_c of the centre, and the two angles that fix $\hat{\boldsymbol{n}}$ relative to the plane through the axis and $\boldsymbol{c}$. Without loss of generality the centre may be placed in the xz-plane, $y_c = 0$. The same reduction applies to the SE, since the open and shielded fluxes share these symmetries.

4.9 Orientation: from Euler angles to unit quaternions

A straightforward way to orient the receiving loop is to compose elementary rotations, for instance a rotation β_x about the x-axis followed by a rotation β_y about the y-axis, $\boldsymbol{Q} = \boldsymbol{R}_y\big(\beta_y\big)\boldsymbol{R}_x(\beta_x)$, which yields the normal

$$\hat{\boldsymbol{n}} = \boldsymbol{R}_y\big(\beta_y\big)\boldsymbol{R}_x(\beta_x)\hat{\boldsymbol{z}} = \big(\sin\beta_y\cos\beta_x, -\sin\beta_x, \cos\beta_y\cos\beta_x\big). \tag{4.49}$$

This parametrisation has three drawbacks. Neither angle has a direct physical meaning, since the tilt of the loop with respect to the shield normal is $\arccos\big(\cos\beta_x\cos\beta_y\big)$. The result depends on the order of the elementary rotations, as $\boldsymbol{R}_y\boldsymbol{R}_x \neq \boldsymbol{R}_x\boldsymbol{R}_y$, so the convention must be fixed and carried through consistently. And the parametrisation is singular: for $\beta_x = \pm\pi/2$ the normal equals $\mp\hat{\boldsymbol{y}}$ whatever the value of β_y, the so-called gimbal lock.

A unit quaternion $q = q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k}$, with $|q| = 1$ and $\boldsymbol{i}^2 = \boldsymbol{j}^2 = \boldsymbol{k}^2 = \boldsymbol{ijk} = -1$, represents the rotation over an angle κ about a unit axis $\widehat{\boldsymbol{m}}$ as

$$q = \cos\!\left(\frac{\kappa}{2}\right) + \sin\!\left(\frac{\kappa}{2}\right)\big(m_x\boldsymbol{i} + m_y\boldsymbol{j} + m_z\boldsymbol{k}\big), \tag{4.50}$$

acting on a vector $\boldsymbol{p}$, identified with the pure quaternion $p_x\boldsymbol{i} + p_y\boldsymbol{j} + p_z\boldsymbol{k}$, as $\boldsymbol{p} \mapsto q\boldsymbol{p}q^*$ [10]. The corresponding rotation matrix is

$$\boldsymbol{Q}(q) = \begin{pmatrix} 1-2(q_2^2+q_3^2) & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & 1-2(q_1^2+q_3^2) & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & 1-2(q_1^2+q_2^2) \end{pmatrix}. \tag{4.51}$$

The map $q \mapsto \boldsymbol{Q}(q)$ is a smooth two-to-one covering of $\mathrm{SO}(3)$ by the unit sphere S^3, with q and $-q$ representing the same rotation. Unlike any three-angle parametrisation it has no singular points, and the composition of rotations is the quaternion product, $\boldsymbol{Q}(q_2q_1) = \boldsymbol{Q}(q_2)\boldsymbol{Q}(q_1)$.

Combined with the invariance of Chapter 4.8, quaternions yield a natural parametrisation of the loop orientation. Since only $\hat{\boldsymbol{n}}$ matters, it suffices to take, for each normal, the shortest rotation that carries $\hat{\boldsymbol{z}}$ onto $\hat{\boldsymbol{n}}$. Its axis $\widehat{\boldsymbol{m}} = (\hat{\boldsymbol{z}} \times \hat{\boldsymbol{n}})/\,|\hat{\boldsymbol{z}} \times \hat{\boldsymbol{n}}|$ lies in the plane of the shield, and its angle is the tilt of the loop with respect to the shield normal. With the normal in spherical coordinates,

$$\hat{\boldsymbol{n}} = (\sin\theta\cos\psi, \sin\theta\sin\psi, \cos\theta), \quad 0 \le \theta \le \pi, \tag{4.52}$$

with θ the tilt and ψ the azimuth of the tilt direction, this rotation is

$$q(\theta,\psi) = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(-\sin\psi\,\boldsymbol{i} + \cos\psi\,\boldsymbol{j}), \tag{4.53}$$

or, directly in terms of the normal and free of trigonometric functions,

$$q = \frac{1 + \hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{n}} + (\hat{\boldsymbol{z}}\times\hat{\boldsymbol{n}})_x\boldsymbol{i} + (\hat{\boldsymbol{z}}\times\hat{\boldsymbol{n}})_y\boldsymbol{j}}{\sqrt{2(1+\hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{n}})}}, \quad \hat{\boldsymbol{n}} \ne -\hat{\boldsymbol{z}}. \tag{4.54}$$

Substitution in Equation (4.51) gives the in-plane axes of the receiving loop, the third column being $\hat{\boldsymbol{n}}$ of Equation (4.52),

$$\hat{\boldsymbol{u}} = \begin{pmatrix} 1-(1-\cos\theta)\cos^2\psi \\ -(1-\cos\theta)\sin\psi\cos\psi \\ -\sin\theta\cos\psi \end{pmatrix}, \quad \hat{\boldsymbol{v}} = \begin{pmatrix} -(1-\cos\theta)\sin\psi\cos\psi \\ 1-(1-\cos\theta)\sin^2\psi \\ -\sin\theta\sin\psi \end{pmatrix}. \tag{4.55}$$

Compared with Equation (4.49), this parametrisation has three advantages. Both parameters have a direct physical meaning: θ is the angle between the loop normal and the shield normal, so that $n_z = \cos\theta$ enters Equation (4.45) directly, and ψ is the direction in which the loop is tilted. The construction does not depend on an order of elementary rotations. And the only remaining degeneracy, at $\theta = 0$ where ψ is undefined, is harmless, since Equation (4.53) then gives $q = 1$ for every ψ and the pose itself is well defined. This pole is the unavoidable coordinate singularity of any two-parameter chart of S^2. When the pose is specified by $\hat{\boldsymbol{n}}$ directly through Equation (4.54) it does not occur at all, and the single excluded normal $\hat{\boldsymbol{n}} = -\hat{\boldsymbol{z}}$ corresponds to a loop turned over, for which any axis in the plane of the shield may be taken, for instance $q = \boldsymbol{i}$.

Whenever Equation (4.49) and Equation (4.55) describe the same normal, the two frames differ only by a rotation about $\hat{\boldsymbol{n}}$, so by Chapter 4.8 they yield the same flux. The quaternion form changes the parametrisation, not the results.

4.10 Range of validity

The third assumption of Chapter 4.1 constrains the pose. By Equation (4.46), $z(\alpha)$ attains its minimum $z_c - R\sqrt{u_z^2 + v_z^2}$, and since the rows of $\boldsymbol{Q}$ are unit vectors, $u_z^2 + v_z^2 = 1 - n_z^2 = \sin^2\theta$. The receiving loop lies entirely in region 3 if and only if

$$z_c - R\sin\theta > c. \tag{4.56}$$

4.11 Shielding effectiveness of an arbitrarily placed loop

With the open-circuit voltage proportional to the flux, the SE of a receiving loop in an arbitrary pose is defined as the insertion loss

$$\mathrm{SE}(\boldsymbol{c}, \hat{\boldsymbol{n}}) = 20\log_{10}\left|\frac{\Phi^{\mathrm{open}}}{\Phi^{\mathrm{shield}}}\right|, \tag{4.57}$$

where both fluxes follow from Equation (4.45) for the same pose, with A_φ from Equation (4.14) and Equation (4.28) respectively. In contrast with Equation (4.17), the SE is no longer a single value per frequency but a function on the four-dimensional pose space of Chapter 4.8.

This definition breaks down for poses in which Φ^{open} vanishes. A clear example follows from Equation (4.45): for a loop centred on the axis, $\boldsymbol{c} = z_c\hat{\boldsymbol{z}}$, the offset term vanishes and $\Phi = R^2 n_z \int_0^{2\pi} g\, d\alpha$, so both fluxes vanish identically when the loop is tilted to $\theta = \pi/2$. Close to such poses the SE is a ratio of two small quantities and reflects the orientation of the loop rather than the performance of the shield. Such poses are therefore reported separately rather than interpreted as high or low shielding.

4.12 Numerical evaluation

Equation (4.45) is a double integral, over the loop parameter α and over the spectral variable λ of Equation (4.28). The inner integral is evaluated for every point of the discretised loop. For the outer integral the trapezoidal rule is used, which for a smooth 2π-periodic integrand converges exponentially in the number of points, so that a modest number of points along the loop suffices.

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